

Function Estimation: How to Estimate the $N_q^{\max}(\rho)$ Function

1 What is "Function Estimation"?

In a classic M/M/1 system, many metrics have known theoretical equations:

- Average number of customers in the system (L)
- Average number of customers in the queue (L_q)
- Waiting time (W, W_q)
- Utilization (ρ)

However, the maximum queue length, $N_q^{\max}$, does not have a theoretical formula. Since the M/M/1 model has infinite capacity, the "max" is not theoretically defined. Nevertheless, in a simulation of duration T , the queue experimentally reaches a maximum value. Therefore, $N_q^{\max}$ is a stochastic quantity that must be estimated using statistical methods. Whatever lacks a formula is estimated using data.

2 Data Collection

For each utilization value $\rho = \frac{\lambda}{\mu}$:

Step 1 – Run multiple simulations For each ρ :

- Execute R runs (e.g., 20).
- Each run must be independent (using a different seed).

Multiple iterations are required because the metric is stochastic; a single simulation is not enough.

Step 2 – Measure per run In each run j , we measure:

$$N_{q,j}^{\max}(\rho) = \max_t Q_j(t)$$

which is the maximum observed queue length in that specific run.

Step 3 – Estimate the expected value We calculate the average across runs:

$$\hat{N}_q^{\max}(\rho) = \frac{1}{R} \sum_{j=1}^R N_{q,j}^{\max}(\rho)$$

This $\hat{N}_q^{\max}$ becomes our "data point" for the regression.

ρ	$N_q^{\max}$
0.05	...
0.10	...
...	...
0.95	...

Table 1: Generated Simulation Data

3 Behavior Observation

Before applying regression, we observe the data behavior:

- At low ρ , $N_q^{\max}$ is small.
- At high ρ (0.8–0.95), it increases significantly.
- As $\rho \rightarrow 1$, the queue "explodes."

This behavior indicates that the increase is **non-linear** and likely related to the factor $1/(1 - \rho)$.

4 Regression Model Selection

- **Model 1: Linear Regression relative to ρ**

$$N_q^{\max}(\rho) \approx a + b\rho$$

- **Model 2: Relationship with theoretical $L_q(\rho)$** Incorporating theory into the model:

$$N_q^{\max}(\rho) \approx a + b \cdot \frac{\rho^2}{1 - \rho}$$

- **Model 3: Non-linear model (Power Law)** The strongest approach for explosive growth:

$$N_q^{\max}(\rho) \approx \frac{A}{(1 - \rho)^B}$$

5 Model based on Theoretical $L_q(\rho)$

Since the theoretical mean $L_q = \frac{\rho^2}{1 - \rho}$ grows explosively as $\rho \rightarrow 1$, we assume $N_q^{\max}$ follows a similar stochastic process. **The Model:** $N_q^{\max}(\rho) \approx a + b \cdot L_q(\rho)$

- **a (Baseline):** The value of $N_q^{\max}$ when theoretical L_q is near zero.
- **b (Scaling factor):** How much more "explosive" $N_q^{\max}$ is compared to L_q .

This is a "theoretically-guided" regression model.

6 The Non-Linear Model (Power-Law)

As $\rho \rightarrow 1$, the queue growth becomes almost vertical. This suggests a power law: $f(\rho) \sim \frac{1}{(1-\rho)^B}$.

- **A:** Height of the curve.
- **B:** Aggressiveness of the explosion. If $B \approx 1$, it resembles the mean; if $B > 1$, the max grows much faster than the mean.

Linearization via Logarithms: The equation $N_q^{\max} = A(1 - \rho)^{-B}$ is non-linear. We transform it:

$$\ln(N_q^{\max}) = \ln(A) + B \cdot \ln\left(\frac{1}{1 - \rho}\right)$$

By setting $Y = \ln(N_q^{\max})$ and $X = \ln\left(\frac{1}{1 - \rho}\right)$, we perform linear regression: $Y = \alpha + BX$, where $A = e^\alpha$.

7 Model Evaluation Criteria

- **Coefficient of Determination (R^2):** Measures explained variability. For queue simulations, $R^2 > 0.90$ is very good, and > 0.95 is excellent.
- **Residuals:** The difference between data and prediction. Ideally, they should be randomly scattered around zero with no patterns (like U-shapes).
- **Visual Comparison:** The curve must follow the scatter points correctly, especially at the "explosive" end.
- **Theoretical Consistency:** It must increase monotonically and explode as $\rho \rightarrow 1$, without giving negative values.
- **Simplicity (Occam's Razor):** The model should be as simple as possible (1-2 parameters preferred) to avoid overfitting.
- **Boundary Behavior:** For $\rho \rightarrow 0$, $N_q^{\max} \approx 0$. For $\rho \rightarrow 1$, $N_q^{\max} \rightarrow \infty$.